Standard errors and geographic dependence

Controls: Standard Errors; Cluster on Historical County / Instrument / MSA / County; Cluster on State; Spatial (Conley); Spatial Cutoff (km); Robust.

1. What a standard error measures. The standard error describes uncertainty in an estimated coefficient under the model's assumptions. It determines the confidence interval and significance stars. Changing the standard-error method usually changes uncertainty without changing the coefficient, provided the estimation sample stays the same. It does not repair an invalid instrument, omitted-variable bias or an inappropriate model. For IV, conventional confidence intervals can also be misleading when the instrument is weak.

2. Robust and clustered errors. Robust allows the size of regression errors to differ between observations, while treating errors from different observations as uncorrelated. Clustering also allows errors within the same group to move together. Cluster on State groups observations by modern state. The number of independent groups, rather than just the number of observations, matters for the reliability of clustered inference.

The other cluster choice depends on the instrument construction. Maximum-density overlap groups observations using the same historical county. County weighted-density overlap groups identical instrument values. County matching and area-weighted population use the modern MSA or county as the group. The label updates accordingly. A cluster containing only one observation provides no additional allowance for geographic dependence between different places.

3. Spatial correlation. Spatial (Conley) allows errors to be correlated between places whose geographic coordinates are within the selected distance. Spatial Cutoff is measured in kilometres. The implementation uses a uniform kernel: pairs within the cutoff receive equal weight and pairs beyond it receive zero weight. A larger cutoff permits dependence over a wider area; it need not increase every standard error. Conley errors are available for ordinary County and MSA regressions, but not for CH models.

4. CH and linked county estimation. Standard State CH uses robust errors. Standard MSA CH permits robust or clustered errors. CH GMM uses asymptotic normal inference without a small-sample covariance correction; nonlinear least squares uses finite-sample corrections.

County-adjusted CH combines uncertainty from the county geography regression and the regional regression. Shared observations are grouped together, so the two stages are not treated as independent. Under MSA clustering, each MSA and its counties share a group; the Robust setting also preserves this necessary grouping. State clustering joins states connected by interstate MSAs. The State specification uses state clusters. This joint calculation uses asymptotic inference and requires an identified county calibration. Schooling returns and the college coefficient remain fixed inputs; their estimation uncertainty is not included.